

- Q.26 Explain frequency to voltage converter with diagram.
- Q.27 What is an accelerometer? Write any four applications of it.
- Q.28 Write any five applications of LVDT.
- Q.29 Explain carbon/resistive microphone.
- Q.30 What are the applications of RTD and thermistors?
- Q.31 What is differential capacitive pick up? Explain its advantages.
- Q.32 How does an electromagnetic pick up work? Explain.
- Q.33 What are the classifications of transducer?
- Q.34 Explain the working piezoelectric pick up.
- Q.35 Explain the working of seismic transducer with diagram.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define transducer. What are the selection criteria of transducer? Write all types of transducers.
- Q.37 Draw the diagram of RVDT. Explain the working principle of RVDT. Give any two differences between LVDT and RVDT.
- Q.38 What is the working principle of inductive transducer? Write any two advantages and disadvantages of it. How does a small displacement can be measured using this?

3rd Sem / Instrumentation & Control Sub : Transducers and Signal Conditioning

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Choose which one is not the characteristic of a transducer?
- It should be highly linear
 - It should be accurate
 - It must be precise
 - The resolution must be very low
- Q.2 The example of an active transducer is ____.
- RVDT
 - Thermistor
 - Photo voltaic cell
 - Bourdon tube
- Q.3 The working principle of LVDT is
- Mutual inductance
 - Self-Inductance
 - Resistance change
 - Capacitance change
- Q.4 A hot wire anemometer measures ____.
- Angular velocity
 - Temperature
 - Acceleration
 - Flow Velocity

- Q.5 Which of the following transducer's output is affected due to dust particles?
- LVDT
 - Potentiometer
 - Capacitive displacement transducer
 - Spring
- Q.6 The rotational displacement can be converted into voltage with the help of
- LVDT
 - Wheat stone bridge
 - RVDT
 - Amplifier
- Q.7 A piezoelectric transducer is a/an
- Active transducer
 - Passive transducer
 - Inverse transducer
 - Both a and c
- Q.8 HWA is an / an
- Resistive transducer
 - Inductive transducer
 - Capacitive transducer
 - Active transducer
- Q.9 RVDT stands of
- Rotary variable differential transducer
 - Resistance variable differential transducer
 - Rotary variable differential transformer
 - Resistance variable differential transformer
- Q.10 VCO converts
- Voltage to frequency
 - Flow into electrical quantity
 - Frequency to voltage
 - None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Expand VCO.
- Q.12 Define microphone.
- Q.13 Inductive micro phone converts _____ into electrical quantity.
- Q.14 A strain gauge converts flow into electrical quantity. (T/F)
- Q.15 What is linearity?
- Q.16 A low pass filter allows all the frequency components above the lower cut-off frequency. (True/ False)
- Q.17 Define re-resolution.
- Q.18 Generally all the sensors are the part of transducers. (True/ False)
- Q.19 Define precision.
- Q.20 What is seismic pick up?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List any five characteristics of transducer.
- Q.22 Explain the working of capacitive micro phone.
- Q.23 Write any five advantages of piezoelectric transducer.
- Q.24 List any five applications of inductive transducer.
- Q.25 Define shaft encoder? What is the working principle of it?